Name

Mathematics Stage 5 Core

Linear Equations with an Algebraic Fraction

Book 1 Further derivatives of functions
Techniques of integration

Version: 261006

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-EQU-C-01 solves linear equations of up to 3 steps, limited to one algebraic fraction

Solve linear equations involving one algebraic fraction

- Solve linear equations involving one algebraic fraction using up to 3 steps

Linear Equations with Fractions

Solving Multistep Equations involving Fractions

Invisible brackets group terms in the **numerator**.

We undo operations in the reverse order of BIDMAS:

1. Undo addition/subtraction outside brackets.
2. Undo multiplication/division outside brackets.
3. Undo addition/subtraction inside brackets.
4. Undo multiplication/division inside brackets.

Example Solve multi-step equations involving fractions

$$\frac{3x}{5} - 6 = 2$$
$$+6 \quad +6$$
$$\frac{3x}{5} = 8$$
$$\times 5 \quad \times 5$$
$$3x = 40$$
$$\div 3 \quad \div 3$$
$$x = \frac{40}{3}$$

$$\frac{4x+3}{6} = -2$$
$$\times 6 \quad \times 6$$
$$4x + 3 = -12$$
$$-3 \quad -3$$
$$4x = -15$$
$$\div 4 \quad \div 4$$
$$x = -\frac{15}{4}$$

Guided Practice

Solve these equations.

a

$$\frac{8x+15}{2} = 8$$

$x = \frac{1}{8}$

b

$$\frac{5x}{2} - 8 = 12$$

$x = 8$

c

$$\frac{4x+15}{9} = 3$$

$x = 3$

d

$$12 = \frac{2(x-10)}{5}$$

$x = 40$

e

$$2\left(\frac{x}{3} - 5\right) = 16$$

$x = 39$

f

$$15 = \frac{3(x+20)}{10}$$

$x = 30$

1 Write the inverse operations done between each line.

a

$$\frac{7j - 12}{9} = 1$$

$\times 9$

$$7j - 12 = 9$$

$+12$

$$7j = 21$$

$\div 7$

$$j = 3$$

b

$$\frac{3r}{5} + 2 = 8$$

-2

$$\frac{3r}{5} = 6$$

$\times 5$

$$3r = 30$$

$\div 3$

$$r = 10$$

c

$$2 = \frac{2x + 4}{7}$$

$\times 7$

$$14 = 2x + 4$$

-4

$$10 = 2x$$

$\div 2$

$$5 = x$$

d

$$8 = \frac{10y}{4} - 7$$

$+7$

$$15 = \frac{10y}{4}$$

$\times 4$

$$60 = 10y$$

$\div 10$

$$6 = y$$

e

$$\frac{3(c - 8)}{5} = 6$$

$\times 5$

$$3(c - 8) = 30$$

$\div 3$

$$c - 8 = 10$$

$+8$

$$c = 18$$

f

$$\frac{9(x + 1)}{3} = 12$$

$\times 3$

$$9(x + 1) = 36$$

$\div 9$

$$x + 1 = 4$$

-1

$$x = 3$$

g

$$10 = \frac{5(x - 4)}{8}$$

$\times 8$

$$80 = 5(x - 4)$$

$\div 5$

$$16 = x - 4$$

$+4$

$$20 = x$$

h

$$1 = \frac{25(x + 6)}{100}$$

$\times 100$

$$100 = 25(x + 6)$$

$\div 25$

$$4 = x + 6$$

-6

$$-2 = x$$

2 Match the sequences of operations to each expression.

a $n \rightarrow \div 4 \rightarrow -3 \rightarrow \times 5$

b $n \rightarrow \div 4 \rightarrow \times 5 \rightarrow -3$

c $n \rightarrow \times 5 \rightarrow -3 \rightarrow \div 4$

d $n \rightarrow \times 5 \rightarrow \div 4 \rightarrow -3$

e $n \rightarrow -3 \rightarrow \times 5 \rightarrow \div 4$

f $n \rightarrow -3 \rightarrow \div 4 \rightarrow \times 5$

A $5\left(\frac{n-3}{4}\right)$

B $\frac{5(n-3)}{4}$

C $5\left(\frac{n}{4}-3\right)$

D $\frac{5n-3}{4}$

E $\frac{5n}{4}-3$

a – C, b – E, c – D, d – E, e – B, f – A

3 Solve these equations.

a $\frac{5x}{9}-9=-4$ 9

b $-\frac{u}{4}+15=8$ 28

c $-\frac{y}{3}+10=17$ −21

d $\frac{8c}{3}+5=-11$ −6

e $-\frac{3x}{4}+5=-7$ 16

f $-\frac{5x}{6}-6=14$ −24

g $17-\frac{2x}{3}=1$ 24

h $2-\frac{6x}{5}=3$ $-\frac{5}{6}$

i $-5-\frac{12x}{7}=3$ $-\frac{14}{3}$

4 Solve these equations.

a

$$\frac{2x-12}{3}=1$$

$\frac{15}{2}$

b

$$\frac{3x+6}{2}=15$$

8

c

$$\frac{-3t-6}{7}=-3$$

5

d

$$\frac{-13-4r}{3}=-15$$

8

e

$$\frac{-9+5x}{2}=-22$$

-7

f

$$\frac{12-x}{3}=5$$

-3

5 Read the example carefully and answer the questions.

$$\frac{3x-6}{7}=3$$

$$\frac{3x}{7}=9$$

$$3x=63$$

$$x=21$$



Substitute $x=21$ into the original equation and show that it is not the correct solution.

Describe the mistake made by the student.

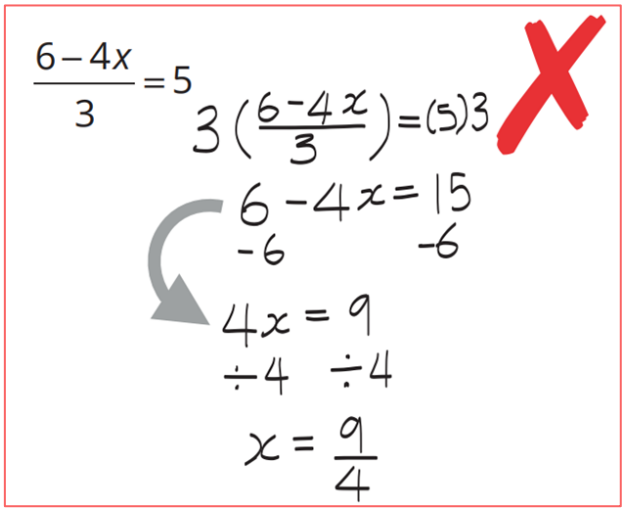
The …… of inverse operations is incorrect. They should have …… first.

Solve the equation correctly.

$x=9$

6 Read the example carefully and answer the questions.

Pablo's work:



What did Pablo forget to do in the step marked with an arrow?

Keep the …… symbol

Solve:

$$\frac{4x-6}{3}=6$$

$x=6$

7 Solve these equations.

a $\frac{2x}{3} - 1 = 7$ — 12

b $\frac{3x}{4} - 2 = 7$ — 12

c $1 = 3 + \frac{x}{2}$ — −4

d $\frac{3b - 8}{2} = 5$ — 6

e $\frac{2x + 2}{3} = 4$ — 5

f $9 = \frac{7m - 8}{3}$ — 5

g $\frac{x - 9}{5} + 4 = 7$ — 24

h $\frac{x - 3}{5} - 8 = -7$ — 8

i $\frac{x + 16}{3} + 5 = 3$ — −22

j $\frac{5 - y}{3} = 2$ — −1

k $\frac{4 - 2t}{6} = 3$ — −7

l $5 = 1 - \frac{4x}{3}$ — −3

m $\frac{2(x - 4)}{3} = 8$ — 16

n $\frac{5(x + 1)}{2} = 10$ — 3

o $\frac{-3(x - 3)}{4} = 0$ — 3

p $\frac{9(x + 10)}{2} = -6$ — $-\frac{34}{3}$

q $\frac{4(2 - x)}{3} = 3$ — $-\frac{1}{4}$

r $\frac{7(x - 9)}{4} = 14$ — 17

8 Solve these three equations.

a

$$-\frac{3c}{4}+5=14$$

-12

b

$$\frac{-3c}{4}+5=14$$

-12

c

$$5-\frac{3c}{4}=14$$

-12

What do you notice?

The solutions are the same.

9 Solve these equations.

a

$$2\left(\frac{x}{3}+1\right)=6$$

6

b

$$5\left(\frac{x}{2}+7\right)=60$$

10

c

$$4\left(\frac{x}{5}-3\right)=-12$$

0

10 Solve these equations.

a

$$\frac{y}{2}+7=19$$

24

b

$$\frac{3y}{2}+7=19$$

8

c

$$3\left(\frac{y}{2}+7\right)=19$$

$-\frac{4}{3}$

d

$$3\left(\frac{y}{2}-7\right)=19$$

$\frac{80}{3}$

11 Consider the equation $\frac{x+4}{5}+3=6$. Solve this equation by:

Subtracting 3 first

Multiplying all terms by 5 first

11

You should have gotten the same answer for both methods above. Which do you prefer?

……

12 Solve these equations.

a $\frac{2}{3}(a-6)=2$ — 9

b $\frac{2(b+7)}{5}=4$ — 3

c $\frac{1}{4}(7-c)=3$ — −5

d $\frac{4}{3}(5-2d)=1$ — $\frac{17}{8}$

13 Solve these equations.

a $-20+\frac{3x}{4}=11$ — $\frac{124}{3}$

b $3(m+5)=18$ — 1

c $4\left(\frac{x}{5}-3\right)=-12$ — 0

d $3(x-2)=10$ — $\frac{16}{3}$

e $6=\frac{3(x-10)}{7}$ — 24

f $\frac{2(p-7)}{3}=\frac{4}{5}$ — $\frac{41}{5}$

g $\frac{3x}{8}-5=1$ — 16

h $\frac{4(x-3)}{6}=\frac{2}{3}$ — 4

i $4(2x+4)=40$ — 3

j $18=2\left(\frac{x}{5}-7\right)$ — 80

k $-4\left(12-\frac{x}{3}\right)=-64$ — −12

l $36=9(k-3)$ — 7

Fraction Equations with Two Unknown Terms

Solving Multistep Equations involving Fractions

If an equation has fractions and more than one pronumeral:

1. Multiply every term on both sides by the denominator.
2. Move all pronumerals to one side.
3. Solve for the unknown using inverse operations.

Example Solve equations with an algebraic fraction and pronumerals on both sides

$$\frac{3x-5}{2}=x$$

1. Multiply $\times 2 \quad \times 2$

$$3x-5=2x$$

2. Move $-2x \quad -2x$

$$x-5=0$$

3. Solve $+5=+5$

$$x=5$$

$$\frac{2x}{3}-5=6x$$

1. Multiply $\times 3 \quad \times 3$

$$2x-15=18x$$

2. Move $-2x \quad -2x$

$$-15=16x$$

3. Solve $\div 16 \quad \div 16$

$$-\frac{15}{16}=x$$

Guided Practice

Solve these equations.

a

$$\frac{7x-12}{3}=x$$

$x=3$

b

$$\frac{3x+4}{5}=x$$

$x=2$

c

$$\frac{3x-4}{5}=3x$$

$x=-\frac{1}{3}$

d

$$\frac{2x}{3}+4=x-1$$

$x=15$

e

$$\frac{2x+4}{3}=3x+1$$

$x=\frac{1}{7}$

f

$$x+\frac{x-1}{5}=3$$

$x=\frac{8}{3}$

1 Solve these equations.

a

$$\frac{4x-40}{8}=3x$$

$x=-2$

b

$$\frac{4x+84}{8}=3x$$

$x=\frac{21}{5}$

c

$$\frac{-6x+54}{4}=3x$$

$x=3$

d

$$\frac{3x+6}{7}=x+2$$

$x=-2$

e

$$\frac{4x-51}{7}=3x$$

$x=-3$

f

$$\frac{-4x+57}{3}=5x$$

$x=3$

g

$$\frac{5x}{4}-6=2x$$

$x=-8$

h

$$\frac{3x}{8}=-5x+26$$

$x=\frac{208}{43}$

2 Read the example carefully and answer the questions.

Example

Solve:

$$x+\frac{2x+1}{7}=4$$

$$7x+2x+1=28$$

$$9x+1=28$$

$$9x=27$$

$$x=3$$

Solve these equations.

a

$$x+\frac{x-2}{3}=6$$

$x=5$

b

$$\frac{2x+3}{5}+2=7$$

$x=11$

c

$$\frac{x-4}{3}-2x=11$$

$x=-\frac{37}{5}$

d

$$\frac{4x+6}{5}-3x=8$$

$x=-\frac{34}{11}$

3 Read the example carefully and answer the questions.

Example

$$x - \frac{x-6}{5} = 6$$
$$5x - (x-6) = 30$$
$$5x - x + 6 = 30$$
$$4x + 6 = 30$$
$$4x = 24$$
$$x = 6$$

Solve these equations.

a $x - \frac{x-2}{3} = 4$

$x = 5$

b $x - \frac{2x+3}{5} = 3$

$x = 6$

c $2x - \frac{x-4}{3} = 11$

$x = \frac{29}{5}$

d $-3x - \frac{4x-6}{5} = 8$

$x = -\frac{34}{19}$

4 **2021 HSC Mathematics Standard 2 Band 5**

Solve:

$$x + \frac{x-1}{2} = 9$$

$x = \frac{19}{3}$

5 **2014 HSC Mathematics Standard 2 Band 4**

Solve:

$$\frac{5x+1}{3} - 4 = 5 - 7x$$

$x = 1$

6 Solve these equations.

a $\frac{x}{6} = \frac{5}{3}$ 10	**b** $\frac{x}{9} = \frac{2}{3}$ 6	**c** $\frac{x}{3} = x - 8$ 12	**d** $3(2x + 3) = 14$ $\frac{5}{6}$
e $43 = 4 - 5(3x - 2)$ $-\frac{29}{15}$	**f** $10 - (3y + 2) = 2y$ $\frac{8}{5}$	**g** $2x - 9 = \frac{x}{4}$ $\frac{36}{7}$	**h** $\frac{x}{2} = x + 5$ -10
i $6m - 4 = \frac{m}{4}$ $\frac{16}{23}$	**j** $4 = 4 - \frac{2n}{3}$ 0	**k** $-\frac{3}{8}x + 10 = 19$ -24	**l** $x + \frac{2x - 4}{4} = 3$ $\frac{8}{3}$
m $x - \frac{x + 5}{6} = 2$ $\frac{17}{5}$	**n** $\frac{2k}{5} - k = 54$ -90	**o** $\frac{5(x - 1)}{6} - x = 1$ -11	**p** $x - \frac{4x + 2}{6} = 5$ 16
q $\frac{5x - 6}{7} = -5x + 22$ 4	**r** $-\frac{3(a - 40)}{12} = 0.5$ 38	**s** $\frac{y + 4}{2} = y + 1$ 2	**t** $x - \frac{14(x + 10)}{7} = 7$ -27

7 Solve

$$\frac{7 - 3n}{6} + 4 = 5(2n + 1)$$